

Disproof of Collatz Conjecture by using O-notation

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ABSTRACT

We give the full disproof of Collatz conjecture using big-O operational logic.

Keywords: O-notation, Collatz function, disproof.

INTRODUCTION

The attempt to solve this problem was done by Akram Louiz in his work [1], since many times the conjecture was unproven [2, 3] – indeed, the big-O notation [4, 5] used to estimate the algorithm notation now can be used in theorem proving despite the relation between complexity classes.

DISPROOF

Let's assume the following for Collatz function $f(n)$:

$$O(f(n)) \leq n.$$

Or, in other words, the length of the sequence of Collatz function is less than the number itself.

Let's consider another case where the number has least number of power of two, $2^m - 1$:

$$O(f(n)) = O(f(2^m - 1)) \approx 3^m > 2^m - 1.$$

The above statement is a contradiction, thus, Collatz function cannot be approximated in the given number of steps.

CONCLUSION

It follows that Collatz conjecture is wrong for any integer number.

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